

30. MEASURING DURABILITY: THE GUIDE

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Knowledgeiswatt

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Durability is the ability to decrease as less as possible performance during prolonged exercise. In a previous post, we saw that durability is not related to traditional endurance physiological or performance markers measured at rested state like VO2max, thresholds' power and 20min max mean power output (Read more [here](#)). So, a higher VO2max or FTP doesn't necessarily mean a higher durability. As a consequence, we can't make assumptions on a given athlete's durability level based on physiological or performance measured at rested state: **we have to measure also durability to have a comprehensive physical evaluation.**

So, how to measure durability?

There are 3 ways:

1. **Standardized test (preferred)**
2. **Record Power Outputs after a certain energy expenditure (kilojoules)**
3. **Power-Heart Rate Decoupling**

1. STANDARDIZED TEST

This is the best solution as, while using approach 2 and 3 we are trying to estimate durability (with some limitation that we will see below), durability's standardized test protocols directly measure this physical capacity.

How to design it?

Durability's standardized testing consists in measuring a physiological and performance marker at rested state (test 1), and then repeating the same measurement after a prolonged fatiguing protocol (test 2) . Using this approach, durability's score will be represented by the % decrease in the chosen physiological/performance marker between before and after the fatiguing protocol.



$$\text{Durability score (\%)} = (B - A) / A * 100$$

where:

- A is the physiological/performance marker measured at rested state (Test 1);
- B is the physiological/performance marker measured after the fatiguing protocol (Test 2);

For example, a rider who has 20 min mean maximal power output of 420 W in a rested state and 400 W after a fatiguing protocol, he/she has a durability score equal to -5% (a lower decrease, -2% for example, would mean a better durability).

When designing a durability standardized test, it is important to consider each of the following points:

1. Choose a proper fatiguing protocol
2. Choose a proper physiological/performance marker on which you want durability to be measured
3. Choose a proper standardized nutrition protocol
4. Testing in one day or two days?

5. Right interpretation of the results

1.1 Choose a proper fatiguing protocol:

One of the common 'mistake' when choosing the fatiguing protocol is considering only energy expenditure, 3000 kj for example, without defining a specific combination of duration and intensity to be used to attain a certain energy expenditure.

In fact,

Work (kj) = Power Output (W) * Time (s) / 1000;

So, the same energy expenditure (kilojoules) can be completed with a shorter and higher intensity fatiguing protocol (2h30min with multiple intervals in zone 4-5, for example) or a longer and lower intensity's one (4 hours in zone 2 in a 5/7 zones model, for example). In this scenario, a recent study (2024) by James Spragg and Colleagues showed that **when kilojoules are matched, higher the intensity of the fatiguing protocol (so lower the duration), lower the durability score.** (Read more [here](#))

So, when designing the fatiguing protocol, you can have in mind a general energy expenditure to target (3000 kilojoules for example), **but then you should pick a specific exercise intensity to be performed for a certain duration.**

Should the fatiguing protocol be standardized using relative or absolute intensity?

It is important to use the 'relative' (% of individual physiological thresholds) and not the absolute (a given power output or heart rate, like 200 W, 5 w/kg or 130bpm) intensity!

This is pivotal to have a proper comparison of durability both between different athletes and when comparing durability's progress over time of the same rider.

Let's understand this better comparing this two fatiguing protocol:

- 3hrs at 250W (absolute intensity)
- 3hrs at 90% of first lactate/ventilatory thresholds (relative intensity).

If we are comparing durability between different riders, if rider A have first threshold at 270 W and rider B at 230 W, A would ride at 92% of first threshold, while B would ride at 108% of first threshold. So, it is very likely that A vs B difference in performance after the fatiguing bout would be influenced by exercising below (A) or above (B) the first threshold. In this scenario, the durability score will be strongly influenced by cyclists' fitness at rested state. Consequently, using such approach can't be considered a true durability assessment.

The same hold true when comparing longitudinally the same cyclist. Let's make this example:

- Test 1 january, beginning of preparation, first threshold 300 W
- Test 2 april, competition period, first threshold 320 W (+20 W improvement)

If we decide to standardise the fatiguing protocol as 3hrs at 250 W, it is very likely that the athlete would perform better after fatiguing protocol in april (test 2) just because he/she would ride at a lower relative intensity and further from first threshold when compared to january (test 1). In fact, even if he/she would be riding below the first threshold in both cases, pedalling at 83 vs 78% of first threshold can still influence fatigue accumulation during the 3hrs at 250 W.

So, my suggestion is sticking with relative intensity (90% of power or heart rate at first threshold for example) when standardising the fatiguing protocol. **With this approach there could be some difference in the absolute power output and so in the total energy expenditure, but I consider this a minor limitation compared to exercising at two different relative intensity.**

Let's go further.

What should be the combination between duration and relative intensity of the fatiguing protocol when testing durability?

While as mentioned above, when kilojoules are matched, on average higher the relative intensity of the fatiguing protocol lower the durability score, a cyclist could be more or less durable according to the characteristics (duration and intensity) of the fatiguing protocol chosen. So different durability types (high intensity durability vs low intensity long duration durability) with different physiological underpinnings may exist. So, I think it make sense to choose a fatiguing protocol specific to the effort of the goal race. For example:

- 4hrs at 90% of first threshold for an ironman
- 3hrs with many efforts around or above the critical power/second threshold for a under 23/continental road cyclist.

1.2 Choose a proper physiological/performance marker

Basically, as the definition of durability is '*the % decrease of physiological/performance between rested state and after a fatiguing protocol*', you are pretty free to choose the physiological/performance marker (time trial, VO2max, lactate threshold, critical power, anaerobic reserve, ...) that you prefer. However, the same James Spragg and Colleagues' work cited above (1), in addition to the importance of duration and intensity of the fatiguing protocol, it also reported that, independently by the fatiguing protocol chosen, some cyclist are more durable when the performance marker is a sprint, while others are more durable when it is a longer effort (12min time trial). **So, durability is also very specific to the physio/performance marker used.** Given this, when choosing the physio/performance marker to test, I would use the same approach described above for choosing the fatiguing protocol: **specificity relative to the goal race.**

For example:

- % decrease of first threshold for an ironman (Fig. 2)

- % decrease of mean power output during a 15s maximal sprint for a road cycling sprinter. (Fig. 3)

Here two specific (both for the fatiguing protocol and the physio/performance marker) durability standardised testing protocols based on all the above considerations.

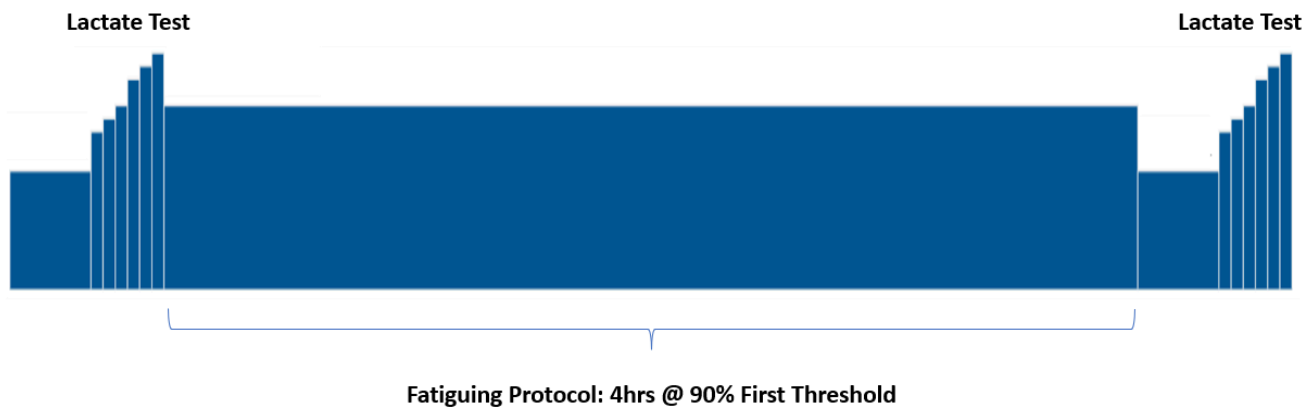


Fig. 2: example of a specific durability's standardised test for an ironman. With lactate test you can measure the first threshold. Image created using Training Peaks.

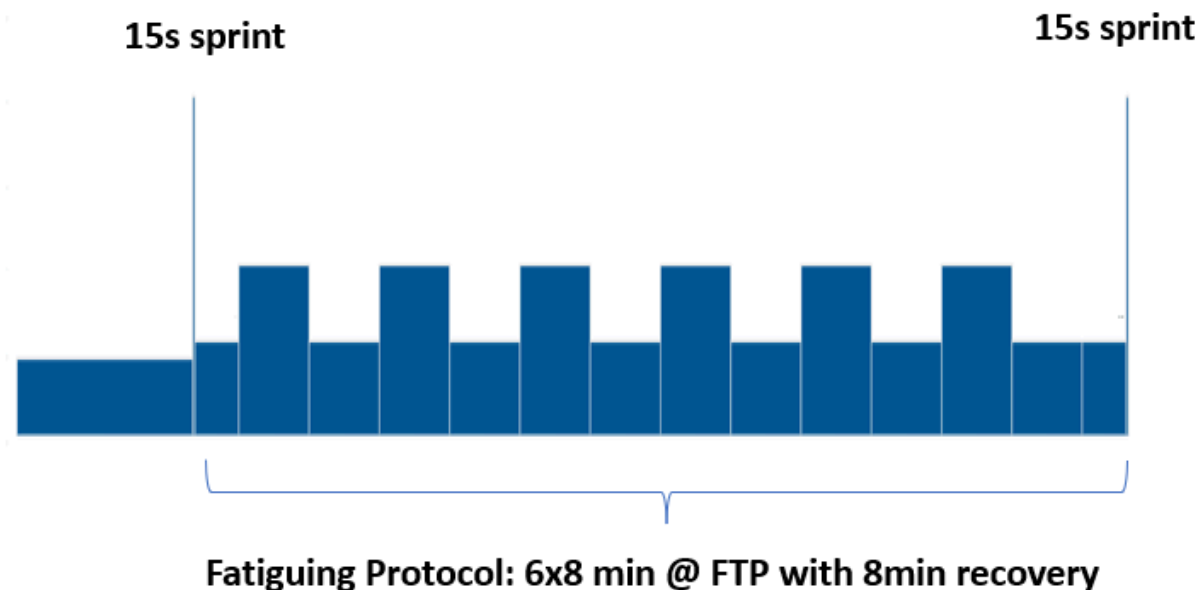


Fig. 3: example of a specific durability's standardised test for a Under 23 Sprinter. Image created using Training Peaks.

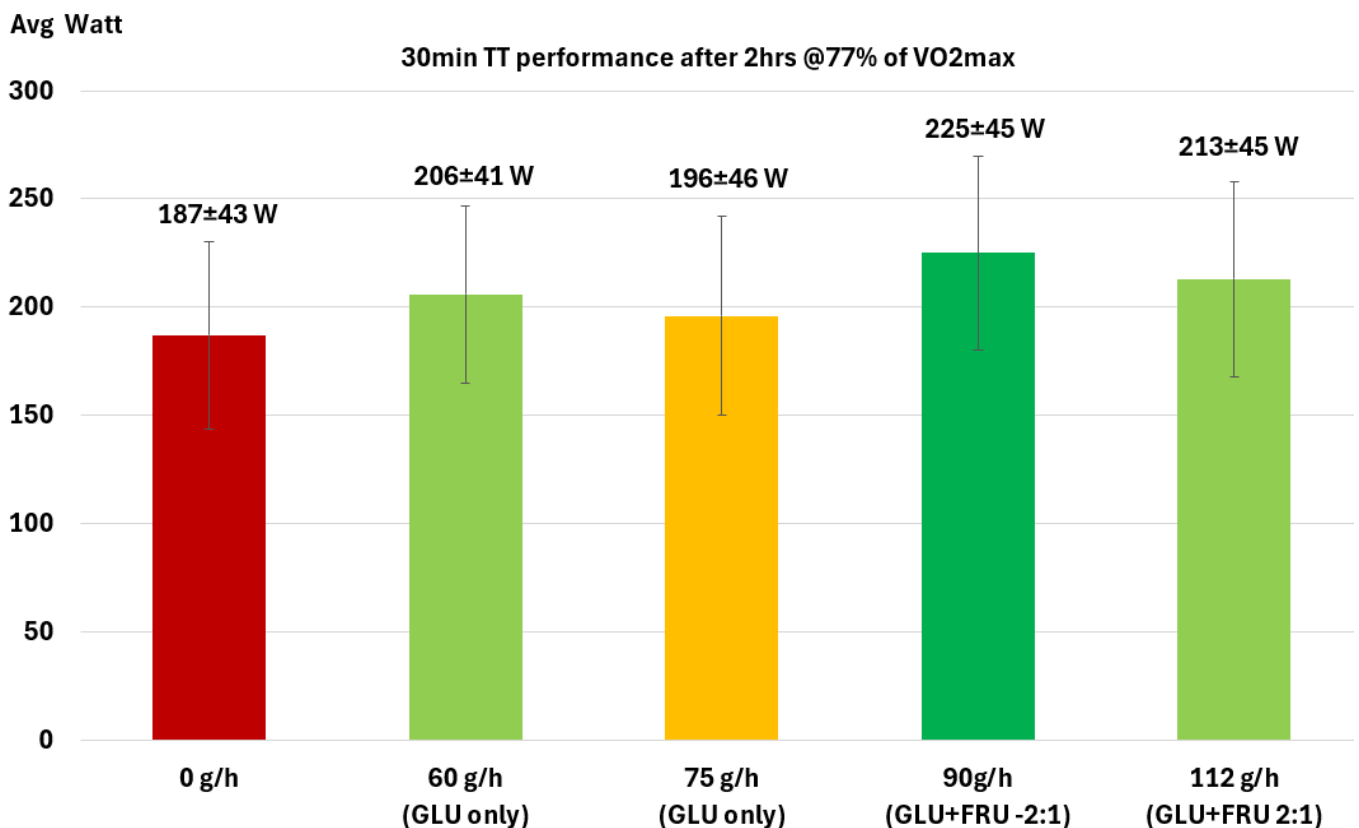
Note that these are only examples and you are free to create different protocols using your ideas and your expertise applied on the principles explained above.

1.3 Choose a proper standardized nutrition protocol

We all know (and several studies showed it) that performance after a prolonged fatiguing protocol is improved when ingesting (exogenous) carbs during the fatiguing protocol itself (2). Why?

In a previous post we saw that durability is in part linked to maintain glycogen (carbs stored in the body) levels during prolonged exercise (read more [here](#)). In fact, while body fat reserves are a virtually unlimited energy resource, body glycogen (carbs) stores are not. Consequently, ingesting exogenous carbs (solid, gels or liquid) during prolonged exercise (read the fatiguing protocol) to use them instead of glycogen to produce energy can delay fatigue and improve durability.

There also seems to be an optimal amount and type of carbs to ingest to optimize performance during prolonged exercise: ~90g/h with a fructose-glucose ratio of 1:2 or higher.



Performance after a fatiguing protocol ingesting different amounts and types of exogenous carbs. GLU is glucose, FRU is fructose. Data from King et al. 2018 (2)

Looking at the graph above, it is very clear that **we can't compare different durability tests performed ingesting different amounts of carbs during the fatiguing protocol.**

Note that glucose's gut transporter is saturated at ~60 g/h and this is probably why '75 g/h GLU only' didn't further improve performance compared to '60 g/h GLU only'. For this reason, the latest scientific studies are trying to verify if using a higher than 1:2 fructose to glucose ratio when exceeding 90 g/h can further improve performance. However, apart from anecdotal evidence from the world of professional cycling, to date (september 2024) there are still no data available suggesting a clear benefit of such approach and one study (read more [here](#)) did not show any glycogen sparing when ingesting 120 g/h with 0.8:1 ratio vs 90 g/h at 1:2 ratio. So we can't say that in general 'more is better'.

So, how standardise exogenous carbs ingestion during the fatiguing protocol when testing durability?

Again, as suggested above when choosing the duration and relative intensity of the fatiguing protocol as well as the physiological/performance marker to test, **I suggest to apply the principle of specificity to the goal race. As performance during a prolonged race is optimized ingesting ~90g/h with a fructose-glucose ratio of 1:2 or higher, this is the exogenous carbs protocol you should use during the fatiguing protocol of durability's standardized test. If during the goal race for some reason you will ingest a lower** (because you are not able to ingest 90 g/h without nausea or because you can't bring with yourself 90 g/h and you have no external assistance) **or higher** (because you are a pro road cyclist and you feel and believe that higher than 120 g/h is better for you) **I suggest to replicate the same protocol during the durability test. In other words, the goal is to test durability with the individualized optimal amounts of carbs that the rider will ingest during the goal race.**

In addition to carbs during exercise, of course it is important to standardise also all the other nutrition and supplements both during the fatiguing protocol and before the test (breakfast?) that can influence the results: water, caffeine, bicarbonate, carbs at breakfast,

1.4 Testing in One Day or Two Days?

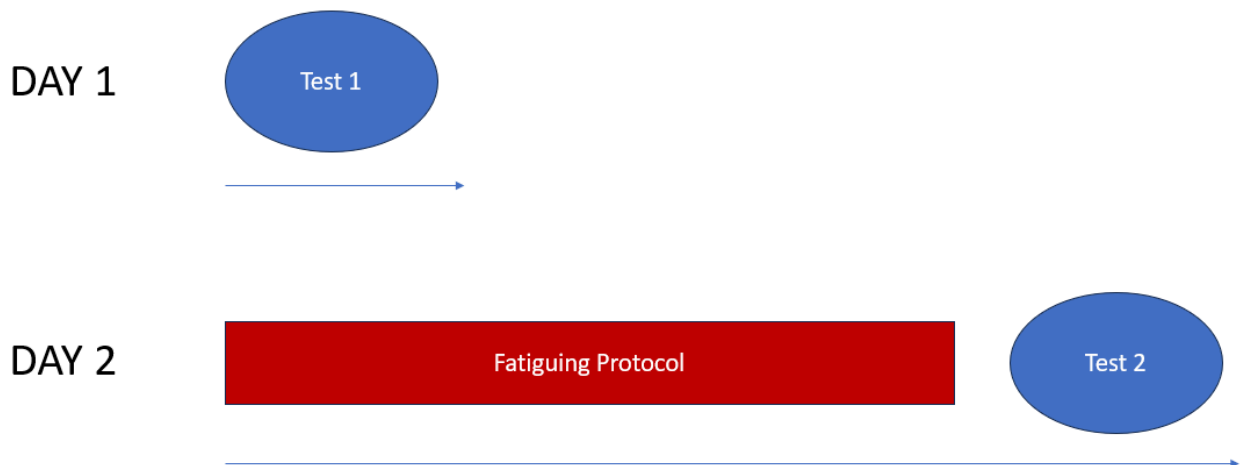
Until now, I describe only durability test protocols where the test at rested state and the test after the fatiguing bout are both perform on the same day.

TESTING DURABILITY IN ONE DAY



However, someone also perform test 1 and test 2 on two different days.

TESTING DURABILITY IN TWO DAYS



Generally, I prefer testing everything on the same day because I think it helps to minimize the day-by-day variability in performance: if you split the test on two days, for any reason an athlete can feel better on one day more than the other and so the durability score could be overestimated (if the athlete feels better on day 2) or underestimated (if the rider feels better on day 1). However there is nothing really wrong performing the two trials on two different days if you are able to standardise the many things that can influence day-by-day variability in performance, especially nutrition and training in the day/s before the tests. Dividing into two different days is usually performed in research setting as it is easier to ask for standardising nutrition and training sessions for some days to non-professional research participants compared to pro athlete who are preparing for an important race.

1.5 Right Interpretation of the Results

It could be quite obvious, but it is better to make it clear: don't be fooled by the results of the durability test!

After finishing the test, you will have three outcomes;

- the physiological/performance marker at rested state (A)
- the physiological/performance marker after the fatiguing protocol (B)
- the durability score

$$\text{Durability score (\%)} = (B - A) / A * 100$$

Both when comparing results between different riders and when looking longitudinally at data of the same rider (after a training block), **the physiological/performance marker after the fatiguing protocol is more relevant than the durability score for race performance(!).**

Let's make an example when comparing two different riders:

	Amateur cyclist	Tadej Pogacar
FTP at rested state	250 W	440 W
FTP after fatiguing protocol	240 W	400 W
Durability score	- 4 %	- 9 %

I don't know the exact Tadej Pogačar's values (the table is just an estimation). In any case, if you consider the numbers in the table, while the amateur cyclist is more durable than Pogačar, who will be faster after the fatiguing protocol? Of course, still Pogačar (400 vs 240 W)

Let's understand the same concept when comparing two tests of the same rider:

	Test 1	Test 2
FTP at rested state	250 W	270 W
FTP after fatiguing protocol	240 W	259 W
Durability score	- 4 %	- 4 %

After a training block between test 1 and test 2, this rider improved his/her FTP both at rested state and after the fatiguing protocol. While durability score remained the same (-4%), when the rider would have been faster at the end of a long race? Of course, Test 2: 259 vs 240 W! So, **when looking how fast a rider will be at the end of a long race, we always have to look at performance after the fatiguing protocol and not at durability score. On the other hand, durability score is very interesting to understand if durability can be an area of intervention to improve performance at the end of a long race, for example adding more low intensity training volume, performing heavy gym strength training or optimizing some nutritional strategy, three things which have been shown to improve durability.**

Until now, we focused on durability's standardised tests. Let's see now the other two methods that can be used to describe this physical capacity: the record power outputs after certain a certain energy expenditure (kilojoules) and the power-heart rate decoupling. As I indicated at the beginning of this guide, these two methods are just estimations with some limitations that I am going to elucidate below. So, my advice is that you can have a look at these two metrics only when you can't perform durability's standardised test, which is the gold standard.

2. RECORD POWER OUTPUTS AFTER A CERTAIN ENERGY EXPENDITURE (KILOJOULES)

This is a widely spread method that many cycling analysis software, (WKO5 for example), can calculate. It consists in comparing the maximal mean power outputs for a given duration recorded during training and races before vs after a certain amount of energy expenditure (kilojoules). It is a retrospective analysis of power meter data for a given period (it is better to select at least 2/3 months to have sufficient data) and so it doesn't require any standardised test.

For example:

- 15s mean maximal power output with <1000 kilojoules spent (this is usually considered as rested state): 1200 W
- 15s mean maximal power output after >3000 kilojoules spent: 940 W
- durability score: -22%

3000kjs is the benchmark usually analysed for professional male road cyclists' durability, as it is approximately the energy expenditure accumulated in races before winning efforts are performed in the last part of the race. However, you are pretty free to choose the kilojoules benchmark more relevant for the athlete you are analysing. For example it could be 1500-2000 kj for women pro road cyclists.

Limitations of this method:

- it doesn't distinguish duration and relative intensity at which kilojoules have been spent, and we saw above that this influences the durability score.
- We are not sure that an all-out effort was performed both at rested state and after the kilojoules benchmark chosen for the performance duration (15s for example) analysed. Gregarious (helpers of the leader) usually perform at their maximum during the first part of the race and then they drop and just pedal to the finish line. So, they almost never perform maximal efforts after 3000 kj. This could underestimate their durability when using this method.

3. POWER-HEART RATE DECOUPLING

The power-heart rate decoupling is a metric representing the heart rate drift to produce a certain power output during the second compared to the first part of a relative long training session/race.

Let's make this example to understand better.

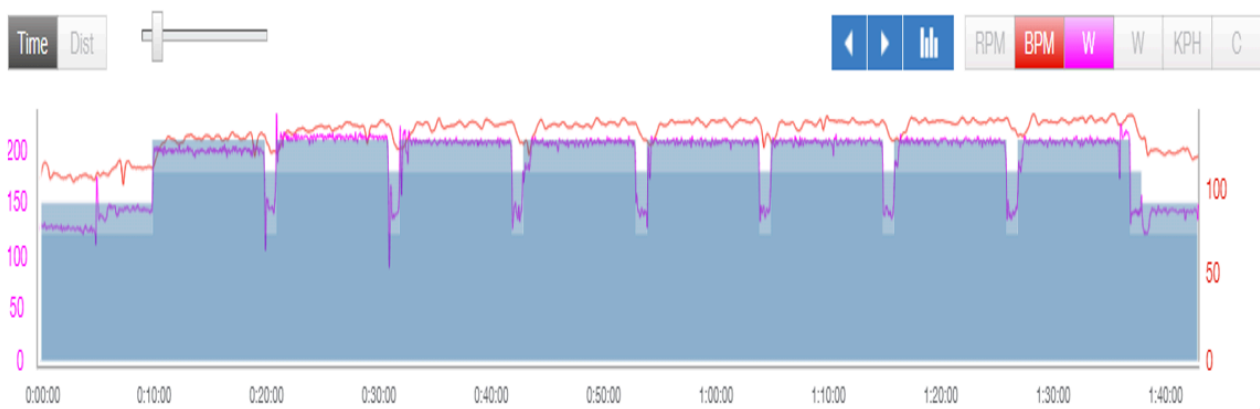


Image from personal Training Peaks account.

This workout represent a 8x10min zone 2 session with 1min recovery (plus warm-up and cool down). During the second 10min @z2 interval, the average power output (purple) was 200 W and the average heart rate (HR) 129 bpm (red). In the last 10min @z2 interval, avg power was 200W and avg HR was 139bpm. So the Power-Heart Rate decoupling score. was -7.7%.

Power-Heart Rate Decoupling Score =

$(\text{W-bpm ratio 2nd part} - \text{W-bpm ratio 1st part}) / \text{W-bpm ratio 1st part} * 100$

My personal advice when using this method is comparing intervals in the same training zone, better if at the same exact power output, so that you can just measure the difference in heart rate.

Is this method valid? Does the power-heart rate decoupling really measure durability?

I am performing a scientific study with my Colleagues of Auckland University of Technology (Dr. Ed Maunder and Dr. Jeff Rothschild) to understand more about this. At the present moment, it looks like ***the power-heart rate decoupling*** can explain about 50% of durability (measured as decrease in power output at first threshold). So the method is 'half-valid'. You can have a look at this metric to have an idea of durability when you can't perform a standardised test but it is not a 100% true durability assessment.

Thanks for Reading! Spread with your Team!

If you have any question, ask in the comments section!

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